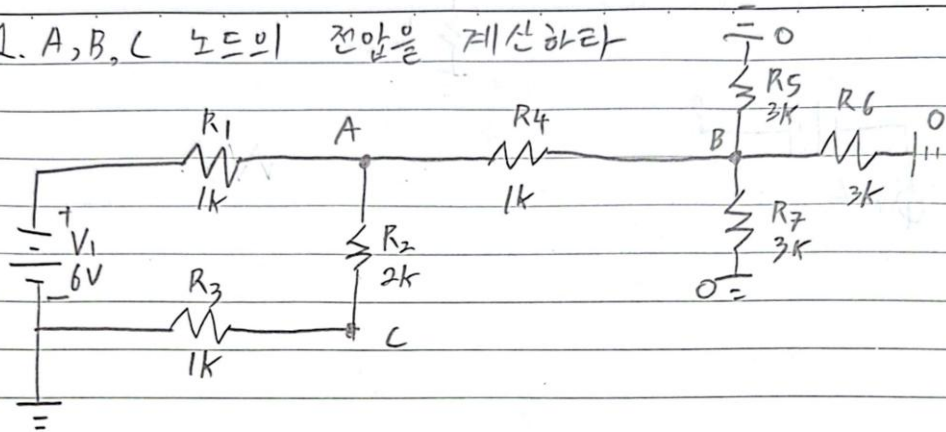


전자회로 해석 및 설계실무

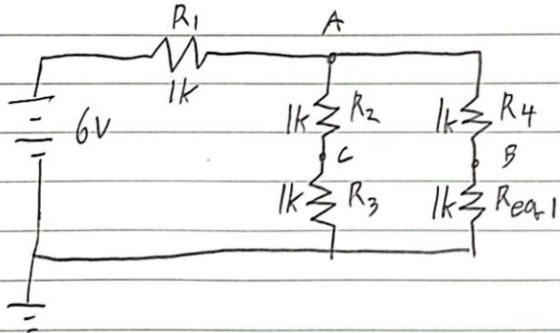
Homework #2

엄민호

1. A, B, C 노드의 전압을 계산하라



$\Rightarrow R_5, R_6, R_7$ 병렬 연결 & 저항 같음 $\Rightarrow \frac{1}{3}R \Rightarrow R_{eq1} = 1k\Omega$
 $\Rightarrow$ 회로를 보기 좋게 다시 그리면



R_2, R_3, R_4, R_{eq1} 의 등가 저항 $\Rightarrow 2k // 2k$

$R_{eq2} = 1k\Omega$

R_1, R_{eq2} 같은 $1k\Omega$ 저항 직렬 연결

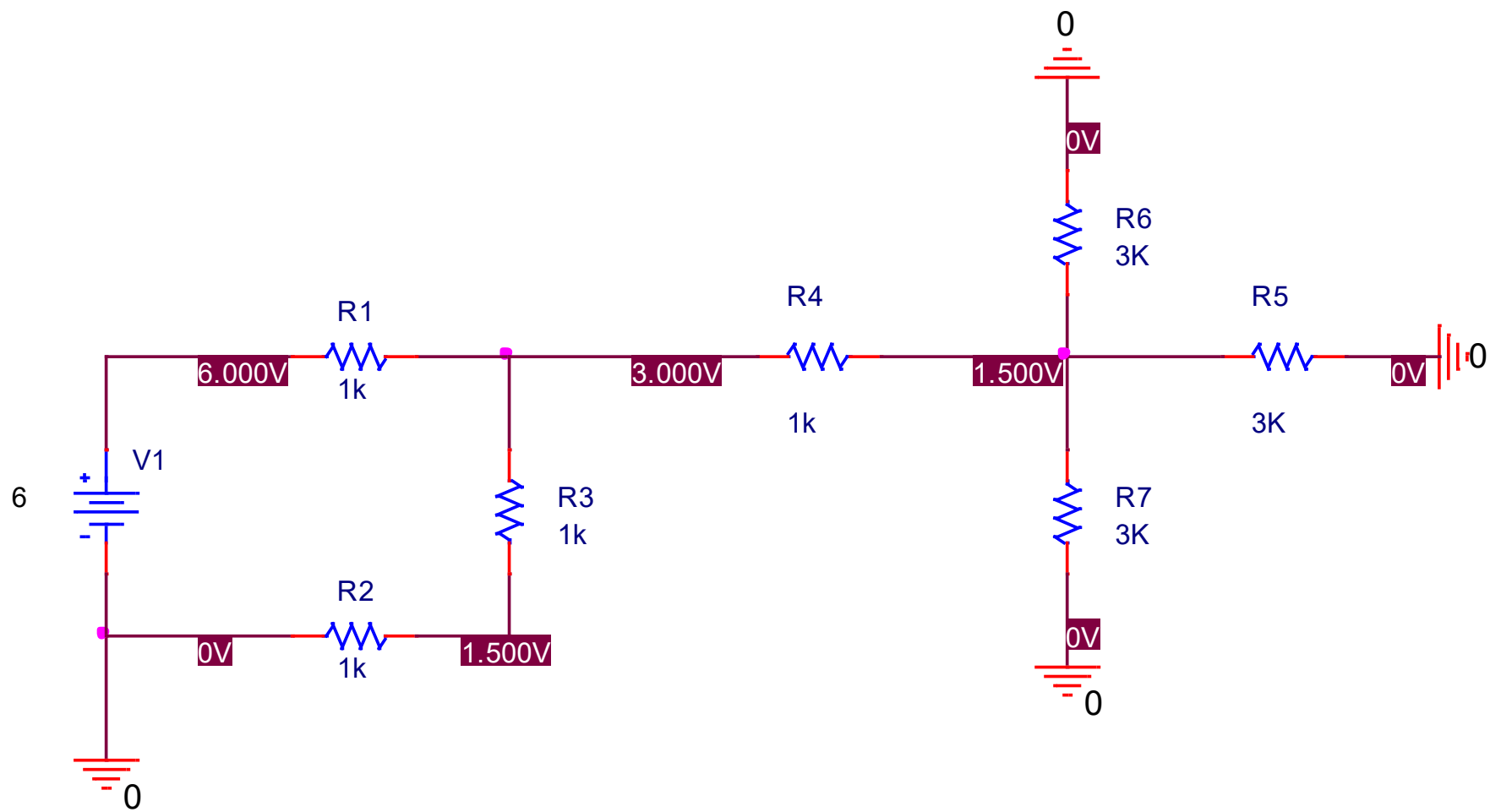
$$\therefore V_A = 6 - V_{R1} = 3V$$

$R_2, R_3, R_4, R_{eq1} \Rightarrow$ Bridge circuit에 저항 같음

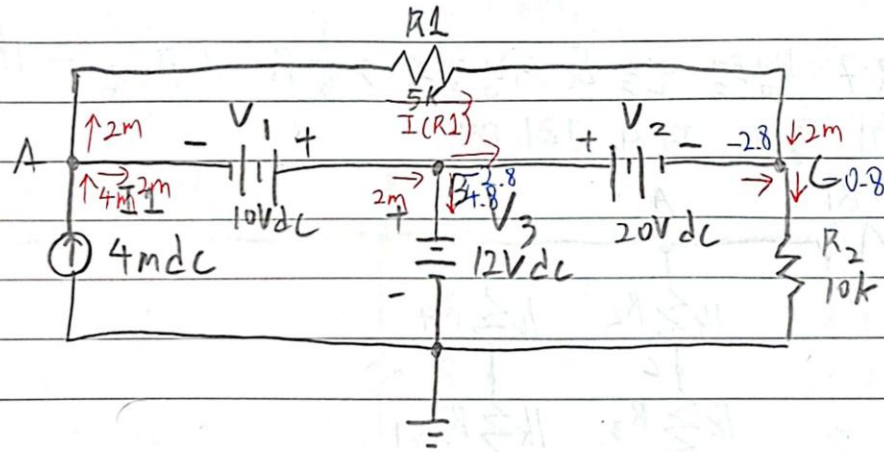
$$V_B = V_C$$

$$V_B = V_A - 0 \times \frac{1}{2} = 1.5V$$

$$\therefore V_A = 3V, V_B = 1.5V, V_C = 1.5V$$



2. A, B, C 노드의 전압과 R1에 흐르는 전류 I(R1)을 계산하고,
source에 해당하는 부품과 load에 해당하는 부품을 구분하시오



$$V_B = 0 + V_3 = 12V, \quad V_A = V_B - 10 = 2V, \quad V_C = V_B - V_2 = -8V$$

$$I_{R1} = \frac{V_A - V_B}{R_1} = \frac{2 - (-8)}{5k} = \frac{10}{5k} = 2mA$$

A, B, C 각 노드 JCL

$$A \Rightarrow I_1 = I_{R1} + I_{V1} \Rightarrow 4m = 2m + 2m \quad I_{V1} = 2m$$

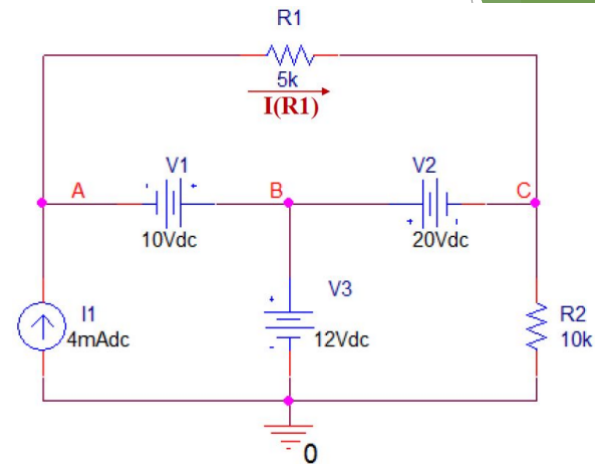
$$B \Rightarrow I_{V1} = I_{V2} + I_{V3} \Rightarrow 2m = -2.8m + I_{V3} \Rightarrow I_{V3} = 4.8mA$$

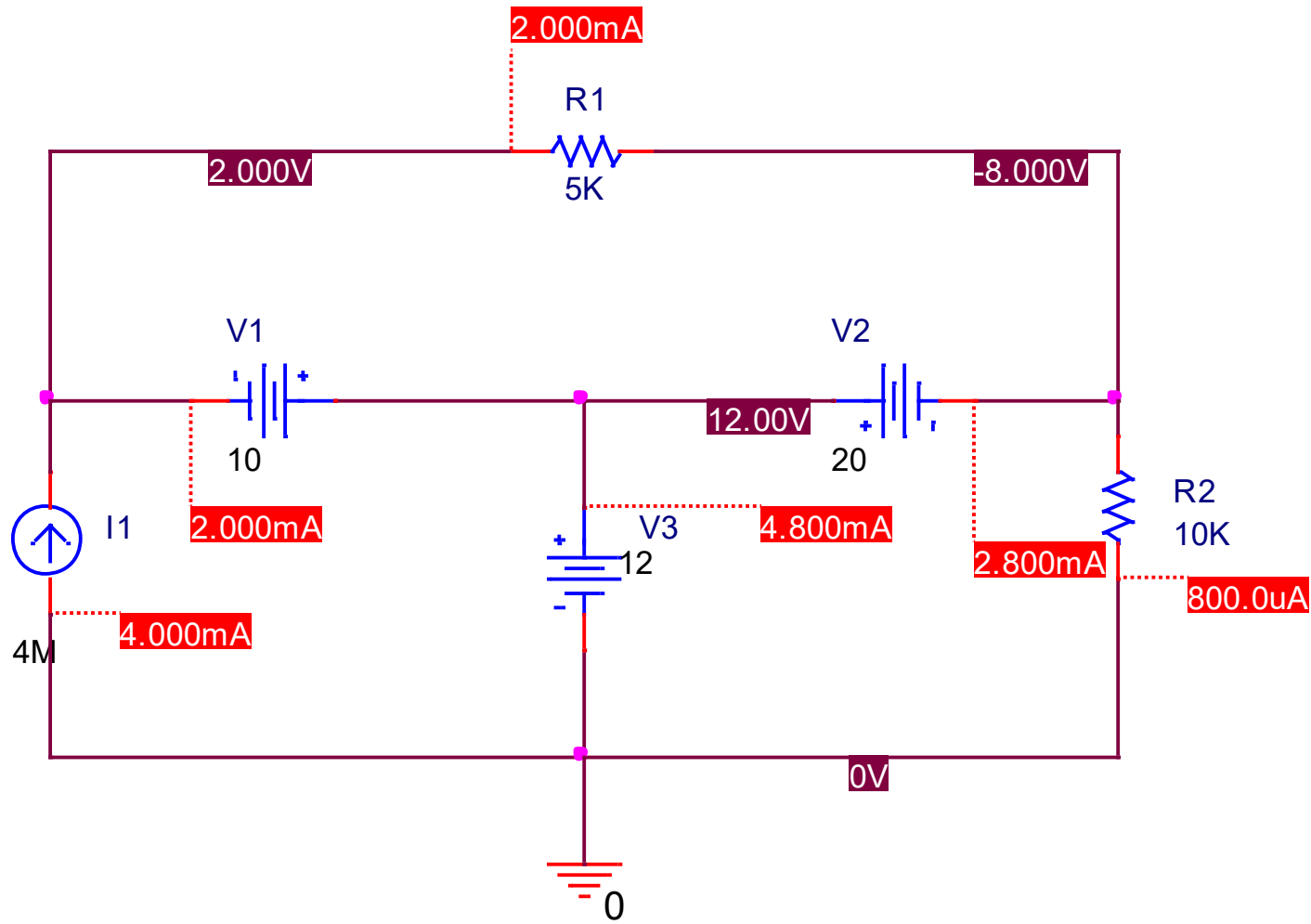
$$C \Rightarrow I_{R1} + I_{V2} = I_{R2} \Rightarrow 2m + I_{V2} = \frac{-8 - 0}{10k} \Rightarrow I_{V2} = -2.8mA$$

∴ source: I1, V1, V2 Load: R1, R2, V3

$$V_A = 2V, \quad V_B = 12V, \quad V_C = -8V$$

$$I(R1) = 2mA$$

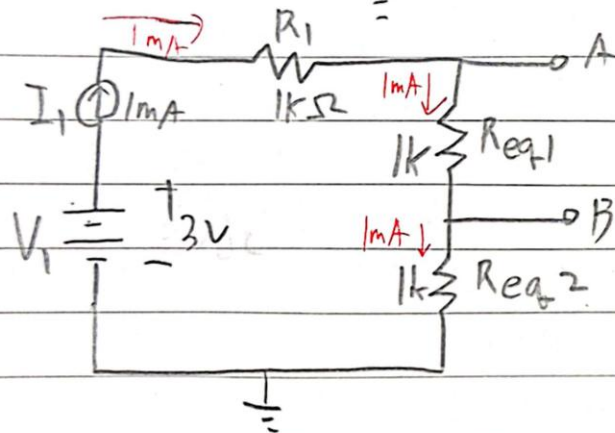
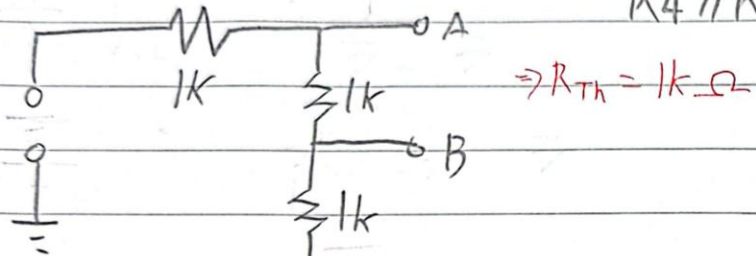




3. A와 B 노드에서의 Thevenin/Norton 등가 회로 구하시오.

$\Rightarrow$ 전압원 short 전류원 open, $R_2 // R_3 \Rightarrow R_{eq1} = 1k$,

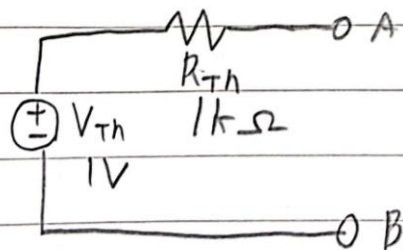
$R_4 // R_5 // R_6 \Rightarrow R_{eq2} = 1k$



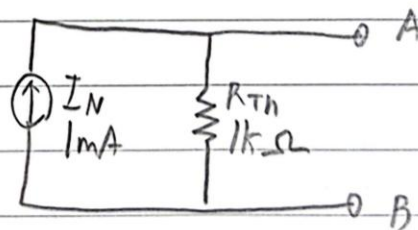
$\Rightarrow V_{Th} = 1m \times 1k = 1V$

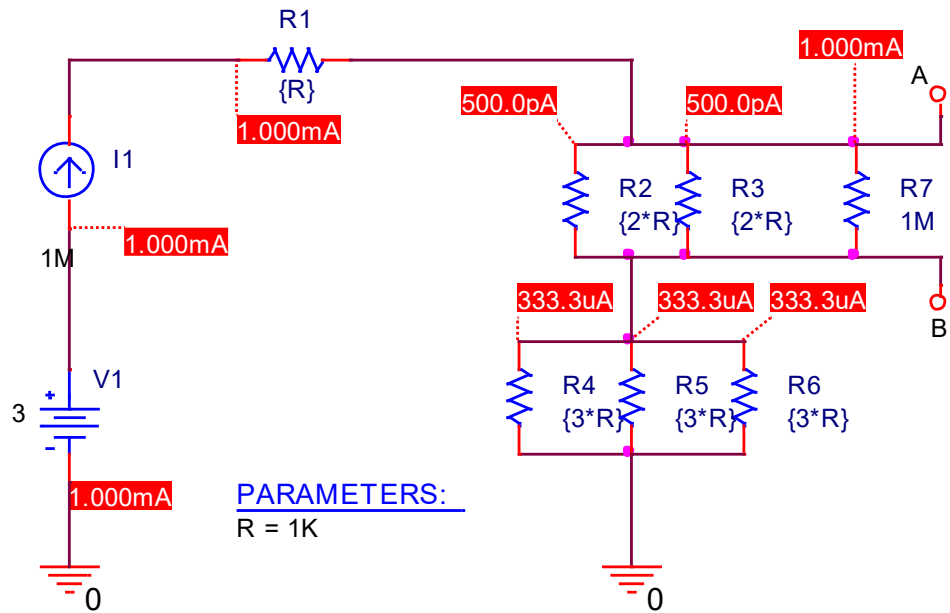
$\Rightarrow I_N = \frac{V_{th}}{R_{th}} = \frac{1}{1k} = 1mA$

Thevenin Equivalent Circuit



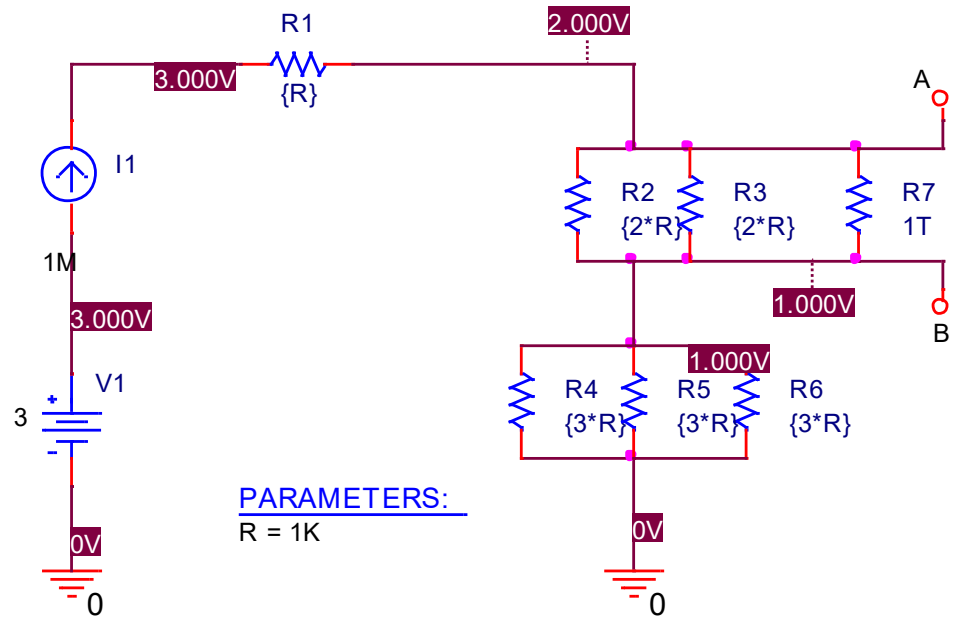
Norton Equivalent Circuit





1m Ω 저항 연결
 $I_{sc} = 1\text{mA}$

$$R_{th} = I_{sc} / V_{oc} = 1\text{k} \Omega$$



1T Ω 저항 연결
 $V_{oc} = 1\text{V}$

